token introspection or by examining the access token itself, which is typically in CWT format. When public key cryptography is used, the token will contain the public key of the client, while for symmetric key cryptography the secret key will be present in COSE-encrypted form, as described in RFC 8747 (https://datatracker.ietf.org/doc/html/rfc8747).

## 13.4 Offline access control

Many IoT applications involve devices operating in environments where they may not have a permanent or reliable connection to central authorization services. For example, a connected car may be driven through long tunnels or to remote locations where there is no signal. Other devices may have limited battery power and so want to avoid making frequent network requests. It's usually not acceptable for a device to completely stop functioning in this case, so you need a way to perform security checks while the device is disconnected. This is known as *offline authorization*. Offline authorization allows devices to continue accepting and producing API requests to other local devices and users until the connection is restored.

> **DEFINITION** ***Offline authorization*** allows a device to make local security decisions when it is disconnected from a central authorization server.

Allowing offline authorization often comes with increased risks. For example, if a device can't check with an OAuth2 authorization server whether an access token is valid then it may accept a token that has been revoked. This risk must